



SpectraPro[®] HRE Echelle Setup and Alignment

Princeton Instruments

ACTON

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3660 Quakerbridge Rd
Trenton, NJ 08619
TEL: 800.874.9789 / 609.587.9797
FAX: 609.587.1970

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System Installation and Alignment

Introduction

The SpectraPro *HRE* Echelle spectrograph incorporates an innovative approach to spectrograph technology, which is ideal for use with CCD and ICCD imaging arrays. The SpectraPro *HRE (HRE)* achieves high spectral resolution over the entire wavelength range of the camera without any moving parts. The system is ideal for both general spectroscopy applications and Laser Induced Breakdown Spectroscopy.

In most cases, your Princeton Instruments camera will be mounted directly onto the *HRE* spectrograph and come pre-aligned and focused at the factory. If your camera and spectrograph were shipped separately, you need to follow the installation instructions provided in **Appendix G**, page 37. Otherwise, continue to "**Focus and Alignment**", page 8.

The *HRE* spectrograph is optimized for cameras with an active image area of 12 x 12 mm or greater. The following Princeton Instruments cameras are designed to mount easily and focus with the *HRE* spectrograph:

- PI-MAX:1K
- PI-MAX:512
- PI-MAX2:1003
- PI-MAX2:512
- PIXIS:1024
- PIXIS:512

In addition, the Photometrics® CoolSNAP™_{K4} camera (with custom spectrograph adapter) can also be mounted to the SpectraPro *HRE*.



Figure 1. SpectraPro *HRE* Spectrograph and Camera

System Components

Standard

The standard system components are:

- HRE spectrograph
- Spectrograph spacer
- Dispersive element (broadband, UV, or deep UV specified at time of order)
- 25 micron aperture
- X-Y stage
- Fiber optic cable (1 meter)
- WinSpec/32 with Echelle Snap-In

Optional

Optional components include:

- Camera (PI-MAX, PI-MAX2, PIXIS, or CoolSNAP_{K4} specified at time of order)
- Additional dispersive element(s)
- Spectrograph entrance shutter
- C-mount (CoolSNAP_{K4} only)
- Fiber optic cable (3 meter)
- 10, 50, or 100 micron aperture
- Scientific Imaging ToolKit™ (SITK™) for LabView

Focus and Alignment

If your camera/spectrograph was shipped as single unit, it has already been focused and aligned at the factory. You will only need to follow these steps to verify that there was no significant movement of the optics during shipment. A standard mercury calibration source is required to perform the focus and alignment procedure. A calibration source with a standard SMA-905 connection works the best, but a glass lamp placed in front of the fiber will also work just fine.

There are two methods to focus the *HRE* spectrograph. There is a coarse adjustment inside the spectrograph, and there is a fine focus adjustment which at one end of the spectrograph.

Coarse Focus

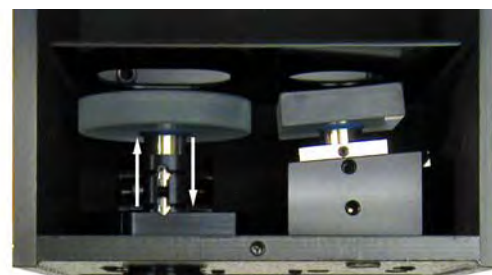
This adjustment will not be necessary when using the camera that shipped with the system, or if the spectrograph was ordered separately and the factory was informed of the camera that will be used with the system. This adjustment may only be used if a camera with a significantly different focal distance is used. Currently, all Princeton Instruments cameras with the standard spectroscopy mount, except for the PI-MAX and PI-MAX2 models, are within the same focal distance and require the spectrograph spacer. An equivalent focal distance for the PI-MAX and PI-MAX2 models is achieved by removing the spacer and mounting directly to the spectrograph.

If you want to switch to a camera with a different focal length, please call the factory for technical support in making this adjustment (see page 46 for contact information). In

some cases, the camera and spectrograph may need to be returned to have the coarse focus adjustment performed.

Fine Focus

1. Power on the camera system and start the application software.
2. Illuminate the CCD with the mercury calibration lamp and take a full image in Focus mode.
3. Identify the 253.652 (or 546.074) nm peak. Refer to the reference images for the dispersive element installed in your spectrograph (see **Appendix A**, pages 17-18).
4. Zoom in on the 253.652 (or 546.074) nm peak into a window approximately 25 pixels x 25 pixels.
5. Click on the autoscale button at the bottom of the software window so that only the brightest pixels can be seen.
6. Make sure the brightest pixel is not saturating the camera. It may be necessary to adjust the intensifier gain and/or the exposure time to bring the intensity below the saturation level. As a rule of thumb, do not exceed 50,000 counts while imaging in the medium analog gain setting.
7. Place the cursor on the brightest pixel of the 253.652 (or 546.074) nm peak. Using the 3/32" hex tool, slightly adjust the fine focus screws (at the end of the spectrograph) while watching the peak intensity.



CW Rotation Pushes the mirror forward
CW Rotation Pulls the mirror backward
CCW Rotation of either screw Does not affect mirror position

Tip: Before rotating either screw clockwise (CW), rotate both screws counter-clockwise (CCW) until the screw heads are fully visible. This does not change the mirror position but allows you adjust in either direction. When a screw bottoms out (or no longer turns) you may have to back the other screw out slightly to continue the adjustment.

Notes:

1. It may be necessary to adjust the intensifier gain and/or exposure time once again if you exceed the 50,000 count limit. When the peak has reached maximum intensity, you are properly focused.
2. Each time you adjust the focus screws, the peak will move slightly and you will have to reposition the cursor so that it sits on the brightest pixel of the peak.

X-Y Stage Alignment

The X-Y stage has two adjustments and a lock-down screw to secure the fiber optic in place when properly aligned. To align the X-Y stage:

1. Using a .05" hex tool, loosen the fiber lock screw.
2. Install the fiber and make sure it is seated all the way into the X-Y stage. Do not tighten the lock screw. You will do this after the X and Y adjustments have been made.
3. Take a full image while running the camera in Focus mode.
4. Identify the 253.652 (or 546.074) nm peak. Refer to the reference images for the dispersive element installed in your spectrograph (see **Appendix A**, pages 17-18).
5. Zoom in on the 253.652 (or 546.074) nm peak into a window approximately 25 pixels x 25 pixels.
6. Click on the autoscale button at the bottom of the window so that only the brightest pixels can be seen.
7. Make sure the brightest pixel is not saturating the camera. It may be necessary to adjust the intensifier gain and/or the exposure time to bring the intensity below the saturation level. As a rule of thumb, do not exceed 50,000 counts while imaging in the medium analog gain setting.
8. Place the cursor on the brightest pixel of the 253.652 (or 546.074) nm peak.
9. Turn either thumbscrew on the side of the stage (see Figure 2) to move the stage in the X direction --- you may need to loosen the opposing screw. While watching the peak intensity, move the stage back and forth (alternating screws) until the peak reaches maximum intensity. It may be necessary to adjust the intensifier gain and/or exposure time once again if you exceed the 50,000 count limit. When the peak has reached maximum intensity, you are properly aligned in the X direction.

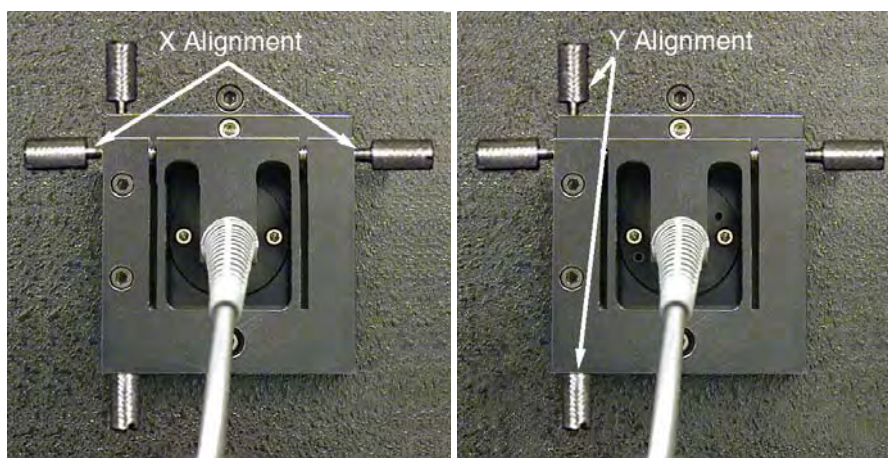
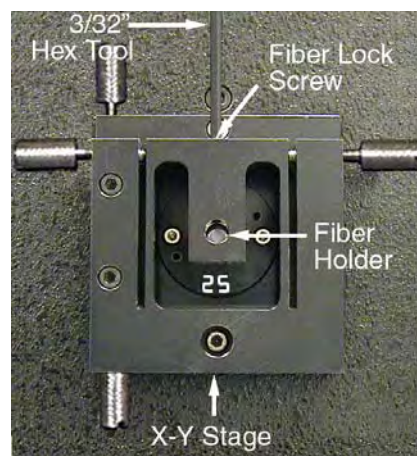


Figure 2. X-Y Stage Adjustment Screws

10. Turn either the thumbscrew on the top or on the bottom of the stage (see Figure 2) to move the stage in the Y direction --- you may need to loosen the opposing screw. Alternate the screws to move the stage up or down. You may need to loosen the opposing screw
11. After completing the X and Y adjustments, tighten the fiber lock screw with a .05" hex tool to secure the fiber in place. Continue to run the camera in Focus mode as in the above adjustments to make sure you do not overtighten the screw. If the intensity drops, you've tightened it too much. Loosen the screw a little until the intensity goes back to the peak level.

Rotational Adjustment

It may be necessary to adjust the rotational alignment of the camera to the spectrograph.

1. Run a full image in Focus mode.
2. Change the cursor to large size and position it on the large 253.652 (or 546.074) nm peak.
3. Loosen the three camera mounting bolts (CoolSNAP_{K4} has four bolts) enough so you will be able to rotate the camera.
4. Using a 3/32" hex tool, back the fine adjust screws away from the rotation adjustment post. When one of these screws contacts the post, it will rotate the camera as you continue to screw it in. For example, tightening down the topmost screw in Figure 3 will turn the PI-MAX or PIXIS camera clockwise (CW). Tightening down the topmost screw in Figure 4 turn the CoolSNAP_{K4} camera counterclockwise (CCW).

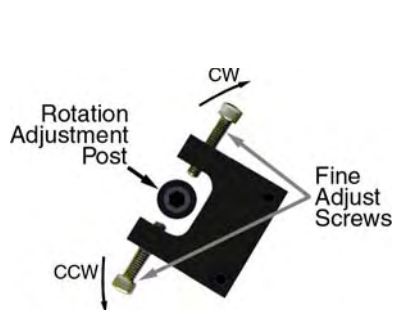


Figure 3. PI-MAX/PIXIS Rotation



Figure 4. CoolSNAP_{K4} Rotation

5. By screwing in the appropriate fine adjust screw, rotate the camera until all the orders are seen on the same row of the CCD as shown in Figure 5. You may have to back off one screw and then screw in the other as you are making the adjustment. When the image is properly aligned, tighten down the mounting bolts to secure the camera in place.

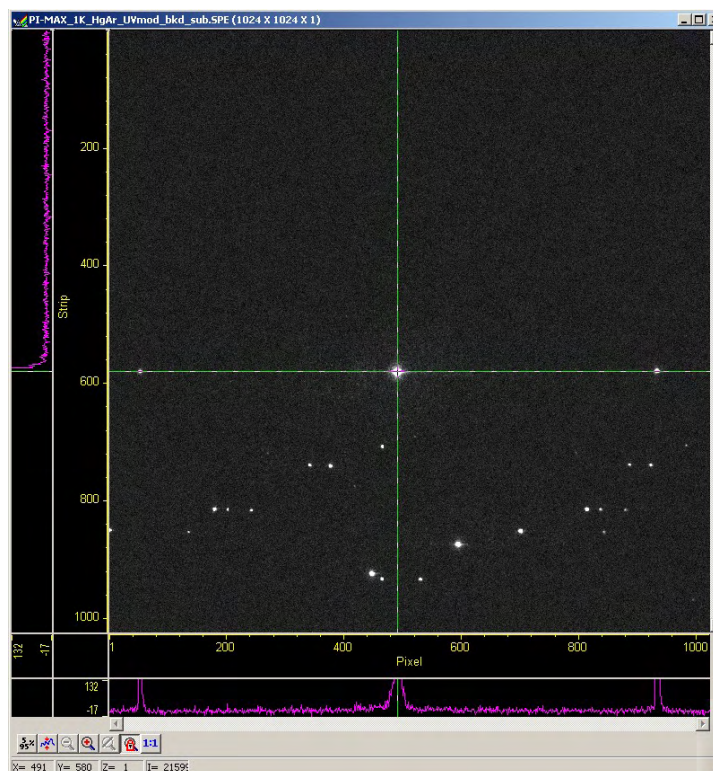


Figure 5. Image showing Rotational Alignment

Wavelength Calibration

Once the camera is properly aligned, the wavelength calibration can be done. To properly calibrate, an image taken with a mercury calibration lamp is necessary. Once calibrated, the information is stored in memory and should only need to be re-calibrated when:

- The system is physically moved to another location
- The dispersive element is removed and re-installed
- A different dispersive element is installed

Calibration of the system requires two points of known wavelength. It is ideal that the separation between the two points be at least one half the size of the CCD (for example, 512 pixels for the 1K, 256 pixels for the 512). Refer to the reference images for the dispersive element installed in your spectrograph (see **Appendix A**, pages 17-18).

1. With the image open, you need to open the **Echelle** snap-in by selecting **Decode Echelle Data...** from the **TOOLS** menu.
2. On the **Echelle Control** dialog box, click on the **Configure...** button to open the **Echelle Decoder Configuration** dialog box (Figure 7).

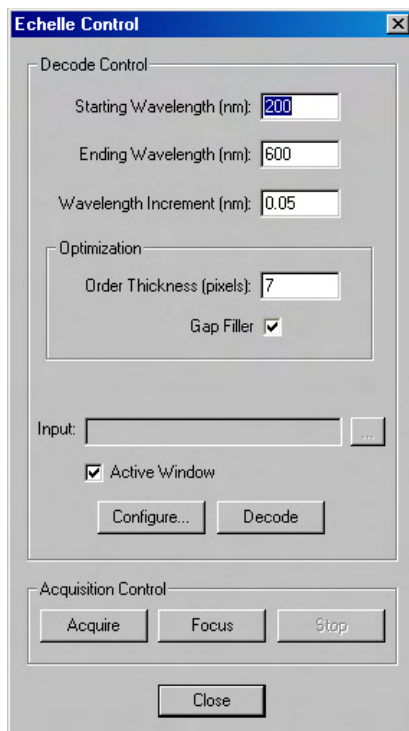


Figure 6. Echelle Control dialog box

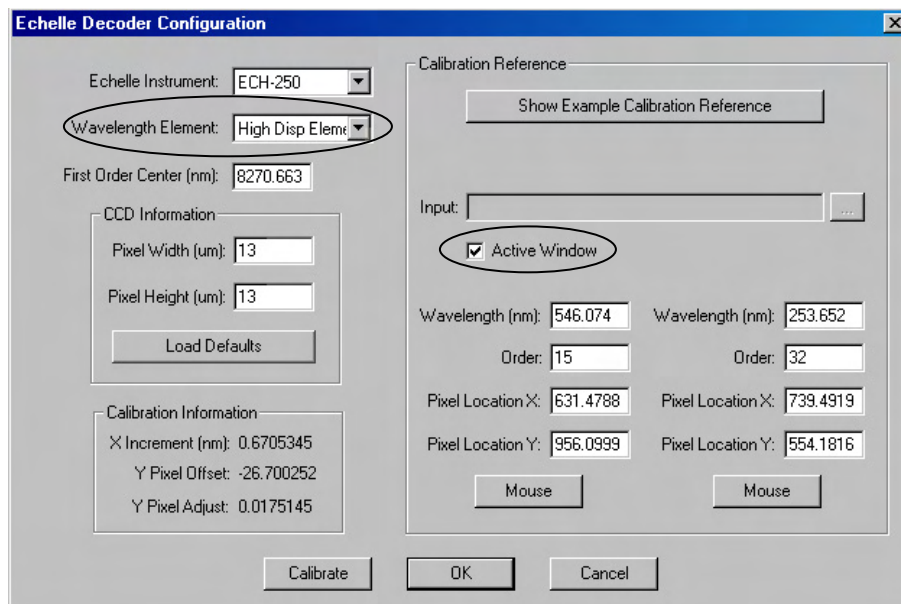


Figure 7. Echelle Decoder Configuration dialog box

3. Choose the correct dispersive element from the dropdown list (see the label on the element installed).
4. Make sure the **Active Window** box is checked.
5. Input the wavelength of the first point you intend to use.
6. Place the cursor on the spot that corresponds to the peak you have just entered and then click on the **Mouse** button (below the column) several times, which will find the center of mass of the given peak and enter the pixel location values.
7. Repeat Steps 5 and 6 for the second point that you intend to use but this time click on the **Mouse** button below the second column.
8. After selecting the two peaks and entering the correct pixel values, click on the **Calibrate** button.
9. Notice that the values for First Order Center, X Increment, Y Pixel Offset, and Y Pixel Adjust will either be entered on the screen or will be changed if they are already there. This indicates that the spectrograph has been calibrated properly.
10. You are now ready to take your data and decode it into a spectrum.

NOTE: If you receive an error message, check the following:

1. Make sure the wavelength you entered corresponds to the correct peak as defined by the reference images in **Appendix A**, pages 17-18.
2. Verify that you have selected the correct dispersive element.
3. Confirm that the peak you have selected has an intensity that significantly higher (at least 500 counts) than the background so it can be discriminated by the software.

Decode Data

In order for spectral data to be displayed by WinSpec software when using the *HRE* spectrograph, it must first be taken in full image mode and then can be decoded by the **Echelle** snap-in.

Take the image you would like to decode or open the file that you want to use.

1. Select **Decode Echelle Data...** from the **TOOLS** menu to display the **Echelle Control** dialog box (Figure 8).

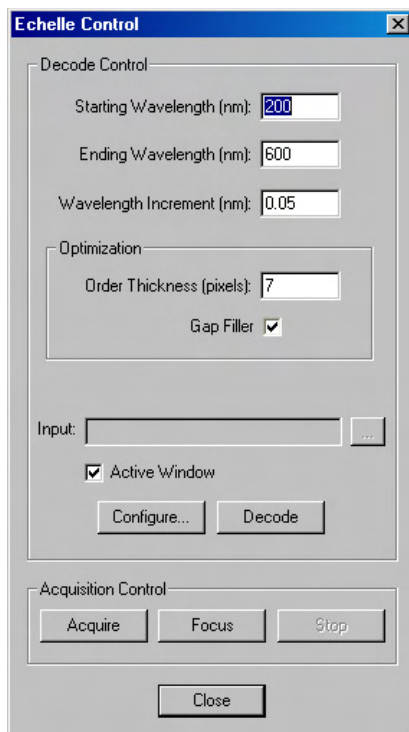


Figure 8. Echelle Control dialog box

2. Select the wavelength range you would like to view by entering the **Starting Wavelength** and the **Ending Wavelength** in the appropriate boxes.
3. Enter the **Wavelength Increment** to set the step size or resolution of the data. It is important to remember that you cannot have more than 65,535 steps in your decoded spectrum. You can calculate the number of steps using the following formula:

$$\text{Steps} = \frac{\text{Ending Wavelength} - \text{Starting Wavelength}}{\text{Wavelength Increment}}$$

For the example above,

$$\text{Steps} = \frac{900 - 200}{0.02}$$

$$\text{Steps} = 35,000$$

4. Check the **Active Window** box if you want to use the window that is currently open in WinSpec or click the browse button (to the right of the Input field) and select the image to be decoded.

- Click the **Decode** button and the spectrum should be displayed on the screen. This file can be saved as a standard WinSpec *.spe file and can be used with any of the post-processing tools, such as Image Math, Peak Find, etc. As can be seen in Figure 9, the resulting image is displayed with intensity in the Y-axis and calibrated wavelength on the X-axis.

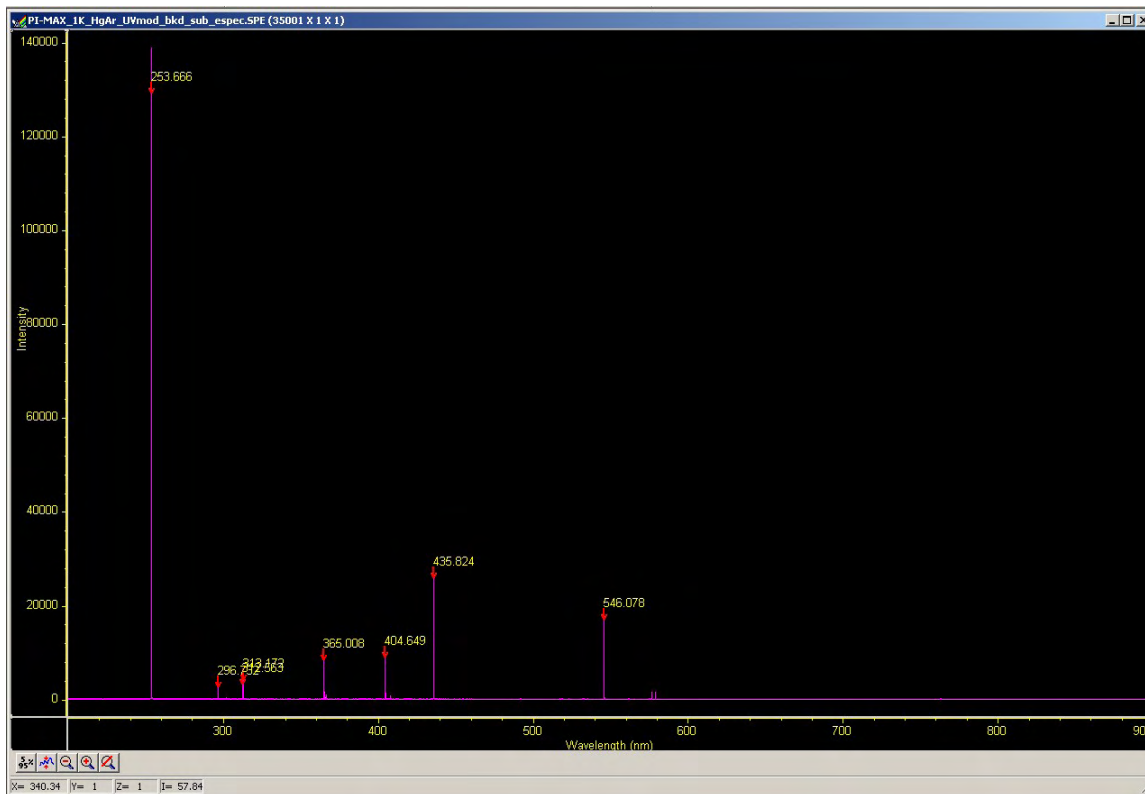


Figure 9. Decoded Spectrum displayed in WinSpec data window

Appendix A

Dispersion Reference Images

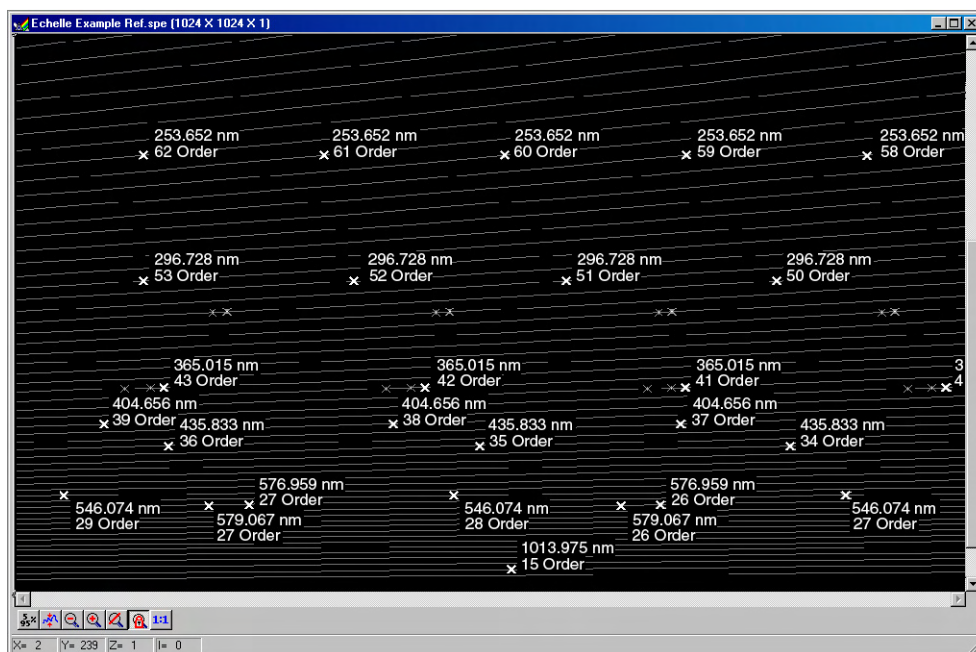


Figure 10. Broadband Dispersion - Hg

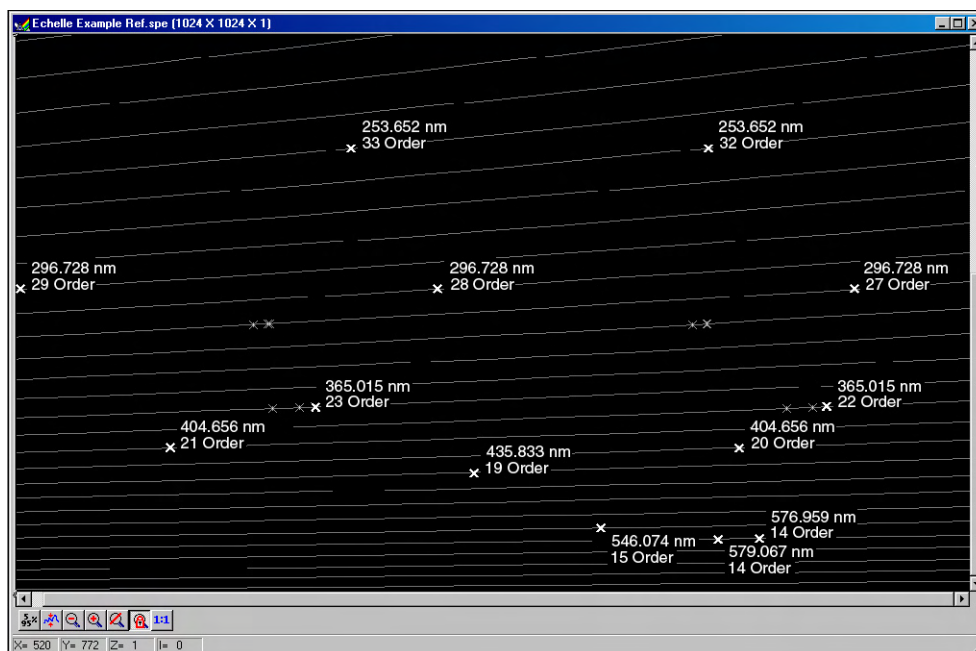


Figure 11. UV Dispersion - Hg

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Appendix B

Specifications and Drawings

SpectraPro HRE Specifications	
Focal Length	250 mm
Aperture Ratio	f/10
Standard Input Aperture	25 μm
Spectral Accuracy	+/- 0.02 nm
Purge Ports	3
Size	5" length x 13.9" width x 6.25" high (127 mm x 353 mm x 158.8 mm)
Weight	13.5 lbs (6.12 kg)

Compatible Cameras			
	Model	Pixel Size	Imaging Area
High resolution	PIXIS:1024	13 x 13 μm	13.3 x 13.3 mm
	PI-MAX:1K (Gen II/III)	13 x 13 μm	13.3 x 13.3 mm
	PI-MAX2:1003 (Gen II/III)	12.8 x 12.8 μm	13.1 x 13.1 mm
	CoolSnap:K4	7.4 x 7.4 μm	15.1 x 15.1 mm
Low resolution	PIXIS:512	24 x 24 μm	12.3 x 12.3 mm
	PI-MAX:512 (Gen II/III)	24.1 x 24.1 μm effective	12.4 x 12.4 mm
	PI-MAX2:512 (Gen II/III)	24.1 x 24.1 μm effective	12.4 x 12.4 mm

Dispersive Elements			
	HRE-DE-VUV	HRE-DE-UV	HRE-DE-BB
Wavelength Range	150 – 350 nm ¹	190 – 600 nm	190 – 1050 nm
Resolving Power²	> 6000	> 6000	> 3000
Resolution (@ 254nm)	0.04 nm FWHM	0.04 FWHM	0.08 FWHM

1 Requires system to be purged with Nitrogen or Argon

2 $\lambda/\Delta\lambda$, using a 13 μm or smaller pixel size

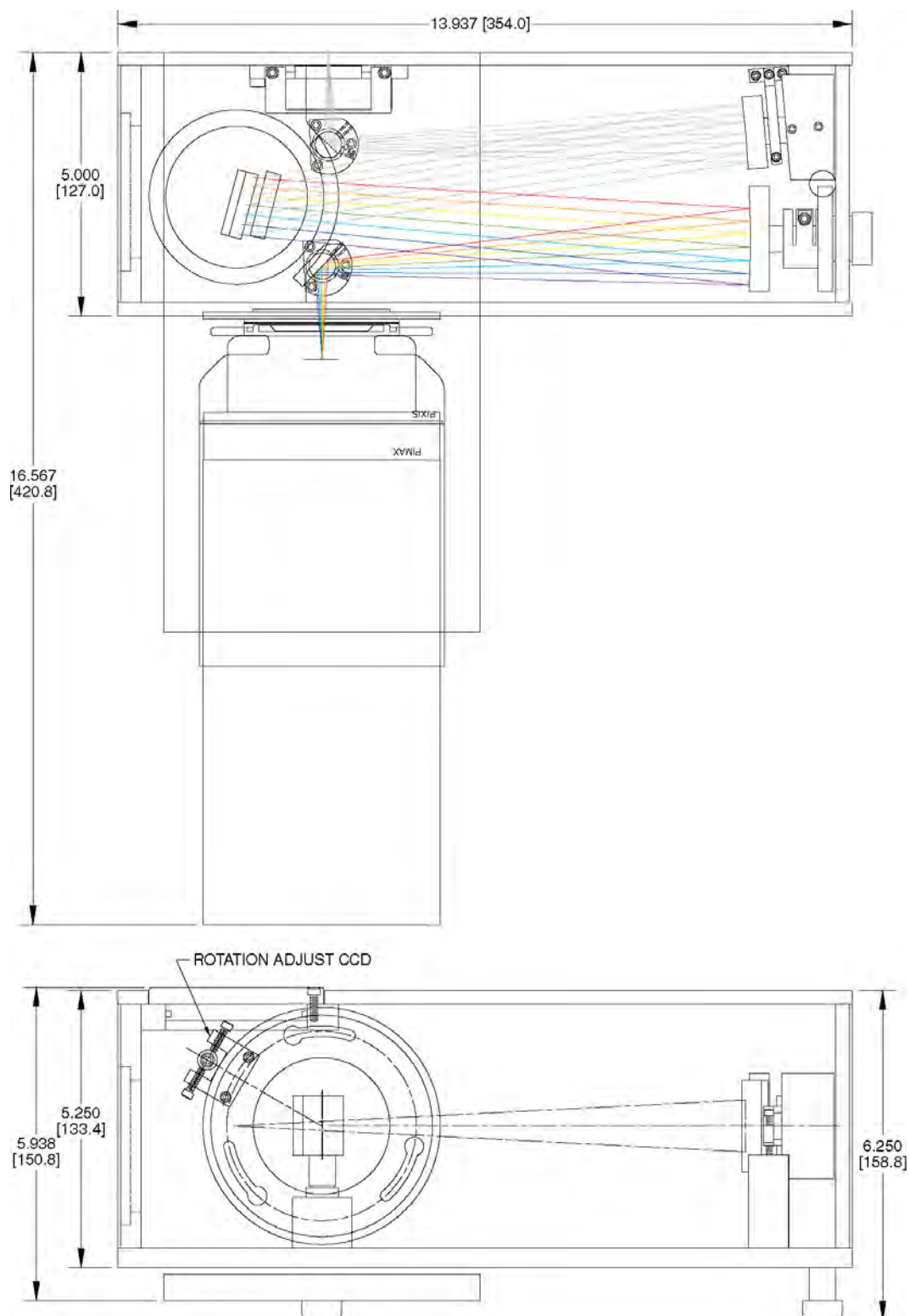


Figure 12. SpectraPro HRE Outline Drawings: Dimensions are in inches [mm]

Appendix C

Installing or Replacing a Shutter

Introduction

Electromechanical shutters typically have a mechanical lifetime of about a million cycles, and depending on usage may need to be replaced. The procedure that follows steps you through testing the currently installed shutter, removing it, and installing a replacement shutter. Replacement shutters can be ordered from PI/Acton: see page 46 for contact information.

Because of the shutter location, there is no need to remove the entrance slit and lose calibration!

Tools List

The following tools are recommended for the following procedure:

Powderless latex gloves

7/64" hex tool (to remove top of spectrograph and remove shutter mount screw)

9/64" hex tool (to remove shutter assembly)

1/16" hex tool (to remove shutter connector)

Shutter Installation

Caution

Use extreme care during these procedures not to directly touch, scratch or allow any tools or components to come in contact with any of the delicate optical surfaces inside the instrument. **Even breathing on these surfaces** will deposit a film which will affect the unit's performance.

1. Using the 7/64" hex tool, remove the (6) top cover screws and set them aside.



Figure 13. Loosening the Cover Screws

2. Lift the cover up off of the spectrograph and set it aside.
3. Using a 7/64" hex tool, remove the light baffle/shutter screw (above the X-Y stage). See Figure 14.

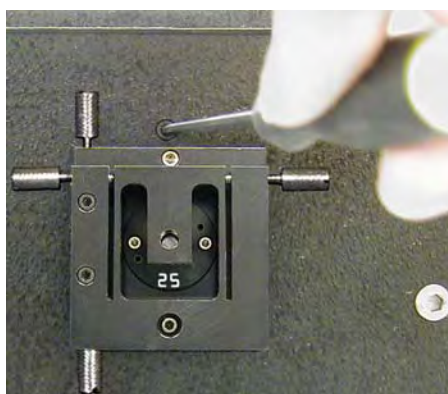


Figure 14. Removing the Light Baffle/Shutter Mount Screw

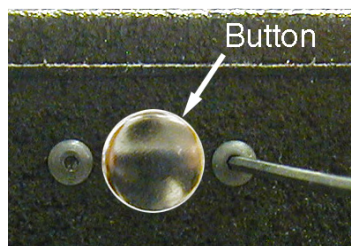


Figure 15. Loosening Socket Head Cap Screws

4. With a small flat screwdriver, remove the button and then use a 1/16" hex tool to remove both plate mounting screws. See Figure 15.
5. Remove the plate from the back wall of the spectrograph.

Note: You may notice some resistance due to the light-blocking foam between the inside wall of the spectrograph and the plate.

6. Slowly slide the light baffle assembly along the back wall upwards at about a 45° angle to remove it from its two support screws (Figure 16). Store the assembly in a safe location.

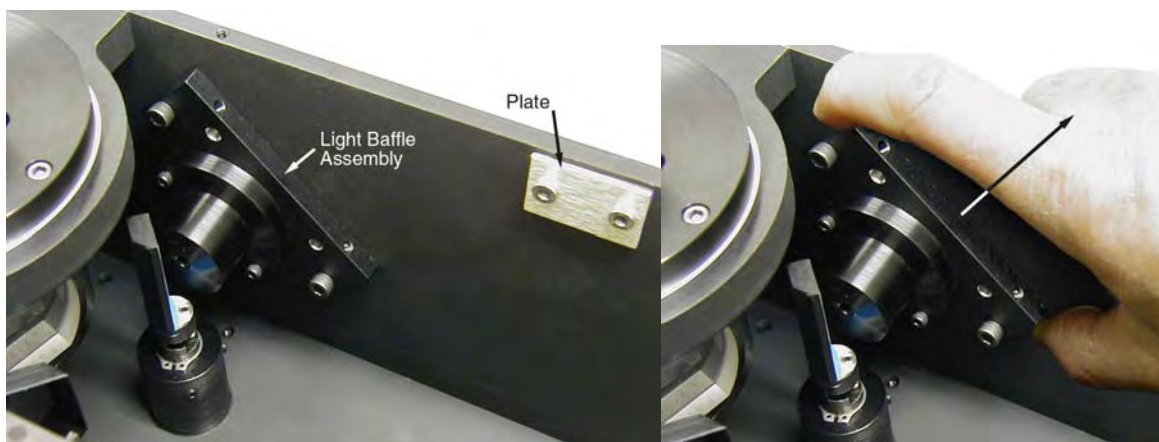
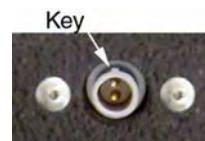


Figure 16. Removing the Light Baffle Assembly



Figure 17. Installing the Shutter

7. Carefully install the shutter assembly and integral connector by aligning the slots and lowering the assembly into position. Keep well away from the optics.
8. Using the 1/16" hex tool, mount the shutter connector. Ensure that the shutter connector key is oriented at the 12 o'clock position and that none of the components/wires are in direct contact with the optics. Snug up (do not over-tighten) both screws.
9. Using the 7/64" hex tool, re-insert and tighten down the light baffle/shutter screw.
10. Replace the cover on top of the spectrograph and snug up all cover fasteners.



Shutter Replacement

Caution

Use extreme care during these procedures not to directly touch, scratch or allow any tools or components to come in contact with any of the delicate optical surfaces inside the instrument. **Even breathing on these surfaces** will deposit a film which will affect the unit's performance.

1. Using the 7/64" hex tool, remove the (6) top cover screws and set them aside.



Figure 18. Loosening the Cover Screws

2. Lift the cover up off of the spectrograph and set it aside.
3. Using a 7/64" hex tool, remove the light baffle/shutter screw (above the X-Y stage). See Figure 19.

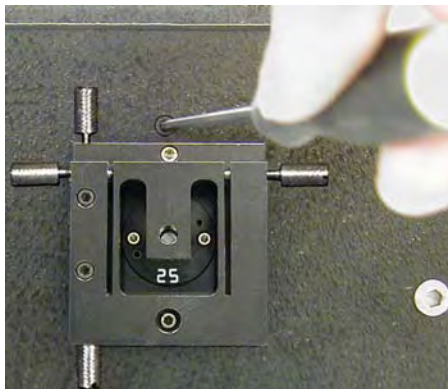


Figure 19. Removing the Light Baffle/Shutter Mount Screw



Figure 20. Loosening Socket Head Cap Screws

4. Using a 1/16" hex tool, remove both connector mounting screws. See Figure 20.
5. Remove the shutter connector from the back wall of the spectrograph.

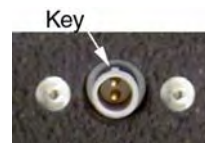
Note: You may notice some resistance due to the light-blocking foam between the inside wall of the spectrograph and the connector mounting plate.

6. Slowly slide the shutter assembly along the back wall upwards at about a 45° angle to remove it from its two support screws. See Figure 21.



Figure 21. Removing the Shutter Assembly/Connector

7. Carefully re-install the replacement shutter assembly and integral connector by aligning the slots and lowering the assembly into position. Keep well away from the optics.
8. Using the 1/16" hex tool, mount the shutter connector. Ensure that the shutter connector key is oriented at the 12 o'clock position and that none of the components/wires are in direct contact with the optics. Snug up (do not over-tighten) both screws.
9. Using the 7/64" hex tool, re-insert and tighten down the light baffle/shutter screw.
10. Replace the cover on top of the spectrograph and snug up all cover fasteners.



The shutter should now be ready for operation. The shutter is normally closed and needs to be activated before light will pass through the instrument.

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Appendix D

Changing the Dispersive Element

Introduction

The SpectraPro *HRE* incorporates a feature that allows you to change the wavelength element and vary the dispersion. The different elements provide a variety of resolution and wavelength ranges for the same CCD. It is no longer necessary to purchase a new camera or spectrograph whenever resolution or wavelength requirements change.

Tools List

The following tools are recommended for the following procedure:

Powderless latex gloves

7/64" hex tool

Element Changeover

Caution

Use extreme care during these procedures not to directly touch, scratch or allow any tools or components to come in contact with any of the delicate optical surfaces on the dispersive element or inside the instrument. **Even breathing on these surfaces** will deposit a film which will affect the unit's performance.

1. Using the 7/64" hex tool, unscrew the three socket head screws holding the dispersive element in place.

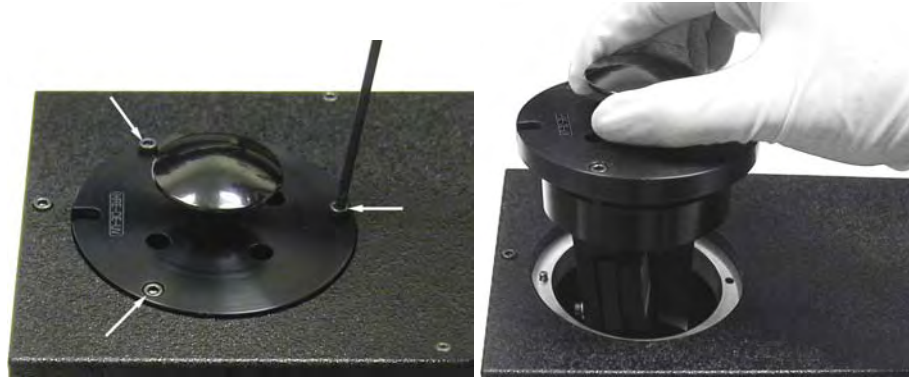


Figure 22. Removing the Dispersive Element

2. Using the knob, lift the element up out of the spectrograph and set the element and screws aside in safe location.
3. Using the 7/64" hex tool, unscrew the three socket head screws mounting the replacement element to its protective container.



Figure 23. Dispersive Element Container

4. Using the knob, lift the element (with screws) out of its container and insert it into the top of the spectrograph. Rotate the element until the notch mates with the locking pin on the spectrograph. This ensures that the element is correctly oriented.
5. Using the 7/64" hex tool, secure the element.



Figure 24. Securing the Dispersive Element

6. Place the element you removed into the container, secure it with the three socket head screws, and store it in a safe location.

Appendix E

Changing the Input Aperture

Introduction

A 25 micron aperture is standard for the SpectraPro *HRE* spectrograph. However, 10, 50 and 100 micron apertures are also available and can be ordered from PI/Acton. Changing from one aperture to another is an easy operation.

Note: Because the X-Y stage is removed and reinstalled during the procedure, you must perform an X-Y stage alignment.

Tools List

The following tools are recommended for the following procedure:

Powderless latex gloves

.05" hex tool

5/64" hex tool

3/32" hex tool

Changing the Input Aperture

1. Using a .05" hex tool, loosen the fiber lock screw (above the fiber) and remove the fiber.

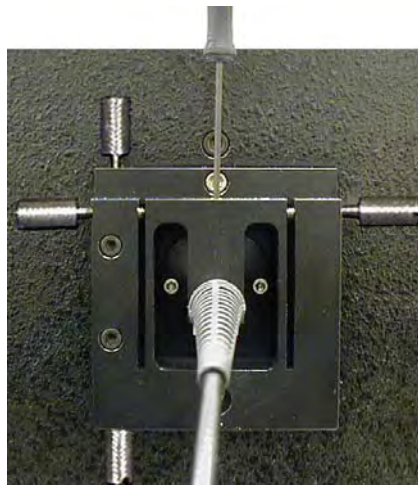


Figure 25. Loosening Fiber Lock Screw

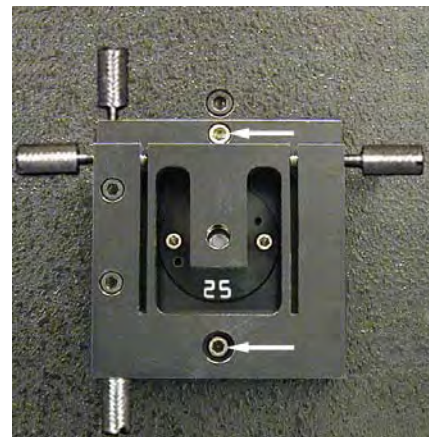


Figure 26. X-Y Stage Mounting Screws

2. Using 3/32" hex tool, loosen two screws mounting the X-Y stage. While supporting the X-Y stage, remove the screws and set the X-Y stage aside.

3. Using a 5/64" hex tool, unscrew the three screws securing the input aperture. Screw two of these into the two threaded holes on the aperture face. You will be using these to remove the aperture.

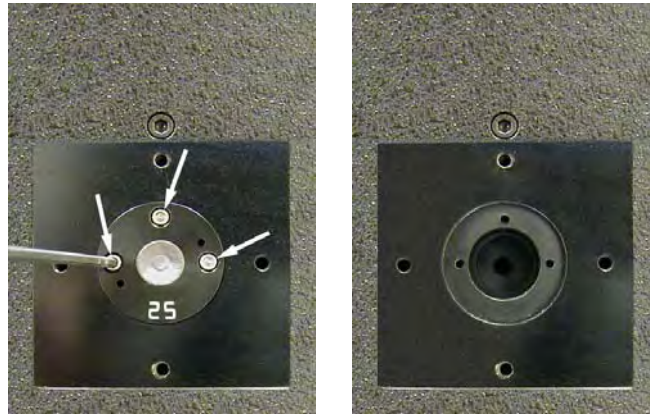


Figure 27. Removing the Aperture

4. Holding the two screws, pull the aperture out, remove the screws, and set the aperture aside.
5. Insert the new aperture.
6. Using a 5/64" hex tool, screw in the three screws to secure the input aperture.
7. Using 3/32" hex tool, remount the X-Y stage.
8. Reinsert the fiber and use the .05" hex tool to tighten the fiber locking screw.
9. Perform the X-Y stage alignment. For more information, see "**X-Y Stage Alignment**", page 10.

Appendix F

Changing the HRE Orientation

Tools List

The following tools are recommended for the following procedure:

Powderless latex gloves

1/8" hex tool

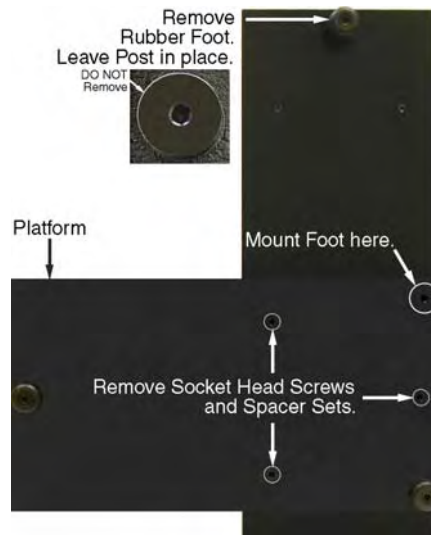
5/32" hex tool

Horizontal (standard) to Vertical (smaller footprint)

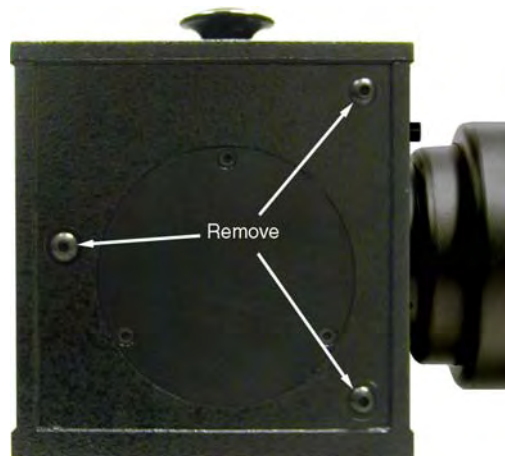


Figure 28. Camera in Horizontal (at left) and Vertical Orientations (at right)

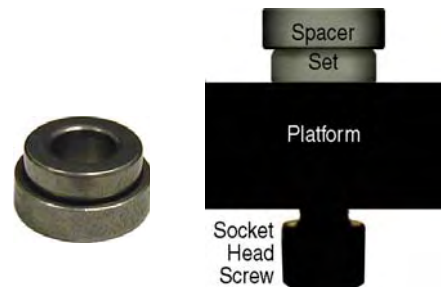
1. First make sure that the detector mounting bolts are firmly tightened down.
2. Rotate the system onto the end closest to the dispersive element. You may need to place a support under the camera (PI-MAX for example).
3. Using a 5/32" hex tool, remove the rubber foot from the focus adjustment end of the spectrograph. Leave the remaining post in place.
4. Mount the foot to the platform.

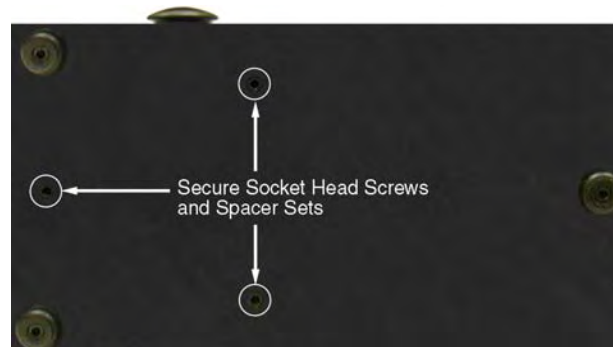


5. Using the same hex tool, remove the three socket head screws that secure the platform. Each screw has a set of two spacers.
6. Rotate the system back to its original orientation and place a support block approximately 1/2" high under the end of the spectrograph closest to the dispersive element.
7. Using a 1/8" hex tool, remove the three (3) button head screws on the end of the *HRE* closest to dispersive element. Set these screws aside.



8. Set the platform on edge and insert the 3 screws from the underside (side with rubber feet).
9. Place the washers on the screws as shown.
10. Carefully position the platform at the end of the spectrograph and use the 5/32" hex tool to screw in one of the socket head screws a couple of turns. Repeat for the other two screws.





11. Now tighten down the three screws.
12. Rotate the system to its vertical orientation.
13. Using the 1/8" hex tool, screw the button head screws (removed in Step 7) into the platform mounting holes.

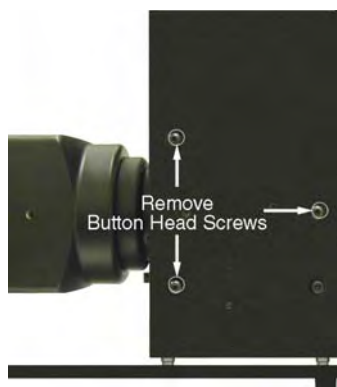


Vertical (smaller footprint) to Horizontal (standard)

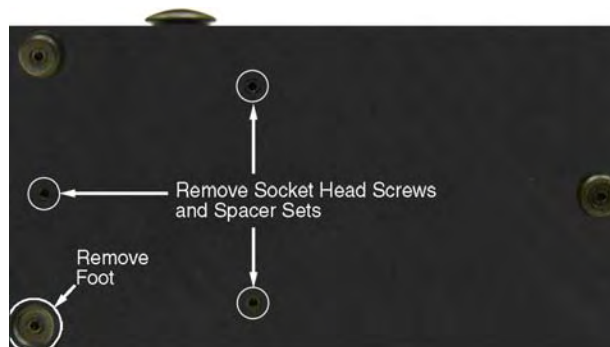


Figure 29. Camera in Vertical (at left) and Horizontal Orientations (at right)

1. First make sure that the detector mounting bolts are firmly tightened down.
2. Using a 1/8" hex tool, remove the three (3) button head screws on the side opposite the dispersive element. Set these screws aside.

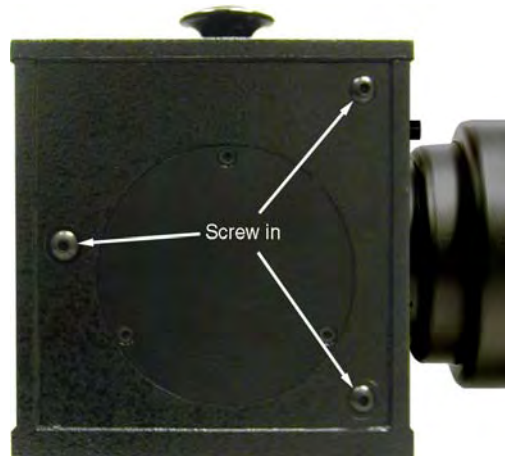


3. Turn the system onto that side. You may need to place a support under the camera (PI-MAX for example).

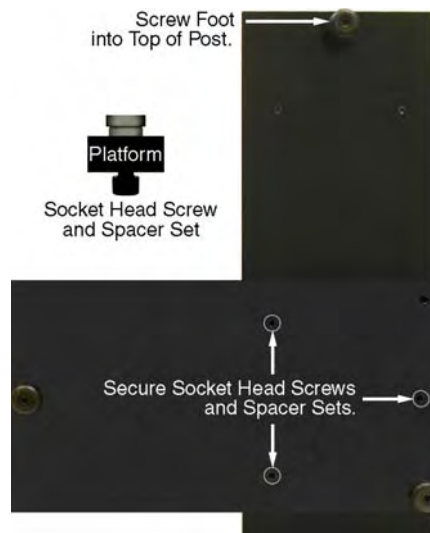


4. Using a 5/32" hex tool, remove the indicated rubber foot.
5. While supporting the HRE, use the 5/32" hex tool to remove the three (3) socket head screw and spacer sets. Set these and the platform aside.

6. Using a 1/8" hex tool, screw in the three (3) button head screws removed in Step 2.



7. Set the HRE on that end. Again, support the camera if necessary.
8. Using the 5/32" hex tool, secure the rubber foot where indicated.



9. With the platform on edge, insert the three (3) socket head screws into the through holes. Then place the spacer sets on the screws as shown.
10. Carefully lift the platform into place and use the 5/32" hex tool to screw in one of the socket head screws a couple of turns. Repeat for the other two screws.
11. Now tighten down the three screws.
12. Rotate the system to its horizontal orientation.



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Appendix G

Camera Installation

Introduction

Typically, a camera is ordered with the SpectraPro HRE, and the system is shipped with the camera already mounted to the spectrograph. If, however, you need to install a camera, remove and re-install an existing camera, or change the mounted camera for another *HRE*-compatible camera, please use the instructions included in this appendix.

Spectroscopy Mount

The camera must be ordered with an *HRE*-compatible spectroscopy mount in order to properly mount the camera to the *HRE*:

- **7389-0003** for PI-MAX and PI-MAX2,
- **7517-0010** for PIXIS[®], or
- **HRE-C-MNT** for CoolSNAP[™]_{K4})

The mounting flange that should be installed on camera is shown below.



Figure 30. PI-MAX, PI-MAX2, and PIXIS Style Mount

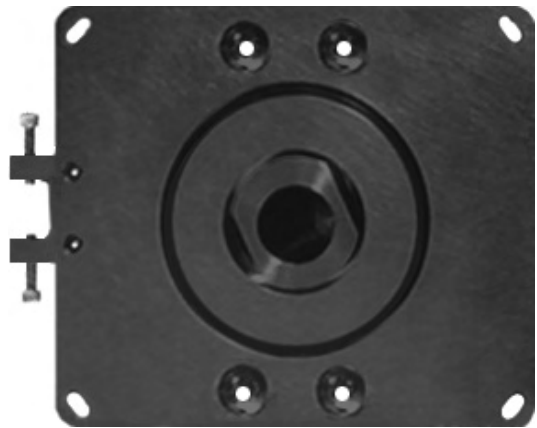


Figure 31. CoolSNAP_{K4} Style Mount

Tools List

The following items may be required to mount the camera to the *HRE* spectrograph:

- Camera with spectroscopy mount
- (3) Hex-head screws (included with camera)
- 1/16" hex tool (user-supplied)
- 3/32" hex tool (user-supplied)

- 7/64" hex tool (user-supplied)
- 5/32" hex tool (user-supplied)
- 9/64" hex tool (user-supplied)
- 3/16" open end wrench (user-supplied)

PI-MAX, PI-MAX2, and PIXIS Installation Procedure

1. Position the spectrograph so its output is facing you. (Figure 32 or Figure 33).
2. Leave the spacer mounted to the spectrograph unless you are mounting a PI-MAX or a PI-MAX2 camera. You must remove this spacer before mounting one of these three cameras. Use a 9/64" hex tool.



Figure 32. Detector Mounting Face with Spacer Mounted to Spectrograph



Figure 33. Detector Mounting Face for mounting PI-MAX or PI-MAX2

3. If you are mounting a PI-MAX or PI-MAX2, use a 5/16" open wrench to screw three (3) bolts into the detector mounting face at the locations noted in Figure 33. These bolts are provided with the camera (or you can use those from the spacer if they are in place). Leave about 3/16" between the mounting face and the back of the bolt heads.
4. If there is rotation adjustment post to the right of the detector mounting face, use a 7/64" hex tool to remove it. Set the post (and its screw) in a safe place -- you will need these if you decide to change over to a CoolSNAP_{K4} at a later date.
5. Rotate the camera so you can easily read the rear panel information (Figure 34).



Figure 34. Camera Orientation with regard to Spectrograph

6. Mount the camera to the bolts on the spacer (PIXIS style camera) or on the detector mounting face (PI-MAX or PI-MAX2).
7. Using the 5/16" open end wrench, tighten the bolts holding the detector's spectrograph flange to the *HRE* but allow for slight movement.
8. If there is a set screw in the mounting hole (to the left of the mounting face) for the rotation adjustment post, remove it with a 1/16" hex tool. Store the set screw or if you removed a rotation adjustment post at the right, insert it into the hole left open: this set screw is used to prevent light from leaking into the spectrograph when the post is not in this location.

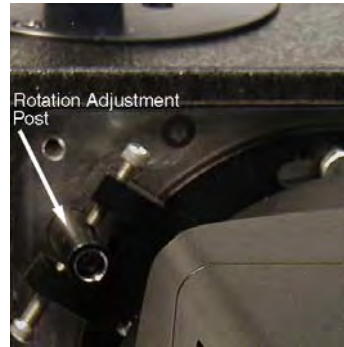


Figure 35. Rotation Adjustment Post

9. Using the 7/64" hex tool, mount the rotation adjustment post at that hole.
10. Using a 3/32" hex tool while looking at a spectrum or target, perform a rotational adjustment. For more information, refer to "*Rotational Adjustment*", page 11.
11. Using the 5/16" open end wrench, tighten the three bolts holding camera's spectrograph mount to the *HRE*.

CoolSNAP_{K4} Installation Procedure

1. If there is rotation adjustment post to the left of the detector mounting face, use the 7/64" hex tool to remove it. Set the post (and its screw) in a safe place -- you will need these if you decide to change over to a PI-MAX, PI-MAX2, or a PIXIS at a later date.



Figure 36. Detector Mounting Face with Spacer Mounted to Spectrograph

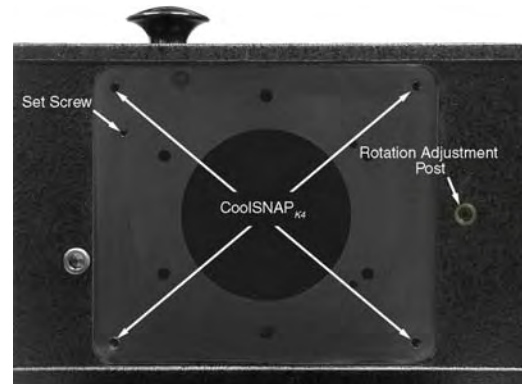


Figure 37. Detector Mounting Face for mounting CoolSNAP_{K4}

2. If there is a spacer, use a 9/64" hex tool to remove it.

3. Using a 1/16" hex tool, remove the set screw from the right side of the detector mounting face. Store the set screw or if you removed a rotation adjustment post at the left, insert it into the hole left open: this set screw is used to prevent light from leaking into the spectrograph when the post is not in this location.
4. Using the 7/64" hex tool, mount the rotation adjustment post.
5. Using a 3/32" hex tool, mount the CoolSNAP_{K4} as shown. Tighten the four (4) corner screws enough to hold the detector in place but allow for slight movement. It needs to be able to move when you make the rotation adjustment.



6. Using the 7/64" hex tool while looking at a spectrum or target, perform a rotational adjustment. For more information, refer to "**Rotational Adjustment**", page 11.
7. Using a 3/32" hex tool, tighten down the four detector mounting screws.

Appendix H

Purging the Spectrograph

Introduction

For specialized applications, the VUV dispersive element can be used to provide coverage from 150 - 350 nm. However, for operation down to 150 nm, you need to flush the HRE with nitrogen or argon.

Procedure

The SpectroPro HRE has three (3) 1/8" NPT purge ports to ensure that each compartment in the spectrograph is fully purged. To illustrate this, Figure 38 shows the spectrograph without its cover and denotes the location of each purge port.

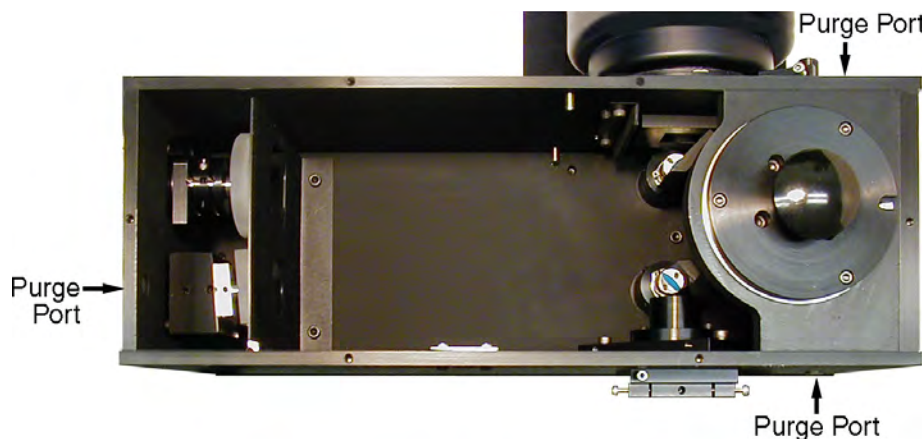


Figure 38. SpectraPro HRE Purge Ports

These purge ports include light reducing devices. The supply line should be opaque to prevent light from entering the instrument. The suggested flow is 20 l/min. Do not let the pressure exceed 0.5psi.

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Warranty & Service

Limited Warranty

Princeton Instruments, a division of Roper Scientific, Inc. ("Princeton Instruments", "us", "we", "our") makes the following limited warranties. These limited warranties extend to the original purchaser ("You", "you") only and no other purchaser or transferee. We have complete control over all warranties and may alter or terminate any or all warranties at any time we deem necessary.

Basic Limited One (1) Year Warranty

Princeton Instruments warrants this product against substantial defects in materials and / or workmanship for a period of up to one (1) year after shipment. During this period, Princeton Instruments will repair the product or, at its sole option, repair or replace any defective part without charge to you. You must deliver the entire product to the Princeton Instruments factory or, at our option, to a factory-authorized service center. You are responsible for the shipping costs to return the product. International customers should contact their local Princeton Instruments authorized representative/distributor for repair information and assistance, or visit our technical support page at www.piacron.com.

Limited One (1) Year Warranty on Refurbished or Discontinued Products

Princeton Instruments warrants, with the exception of the CCD imaging device (which carries NO WARRANTIES EXPRESS OR IMPLIED), this product against defects in materials or workmanship for a period of up to one (1) year after shipment. During this period, Princeton Instruments will repair or replace, at its sole option, any defective parts, without charge to you. You must deliver the entire product to the Princeton Instruments factory or, at our option, a factory-authorized service center. You are responsible for the shipping costs to return the product to Princeton Instruments. International customers should contact their local Princeton Instruments representative/distributor for repair information and assistance or visit our technical support page at www.piacron.com.

Normal Wear Item Disclaimer

Princeton Instruments does not warrant certain items against defect due to normal wear and tear. These items include internal and external shutters, cables, and connectors. *These items carry no warranty, expressed or implied.*

XP Vacuum Chamber Limited Lifetime Warranty

Princeton Instruments warrants that the cooling performance of the system will meet our specifications over the lifetime of an XP detector or Princeton Instruments will, at its sole option, repair or replace any vacuum chamber components necessary to restore the cooling performance back to the original specifications at no cost to the original purchaser. *Any failure to "cool to spec" beyond our Basic (1) year limited warranty from date of shipment, due to a non-vacuum-related component failure (e.g., any components that are electrical/electronic) is NOT covered and carries NO WARRANTIES EXPRESSED OR IMPLIED.* Responsibility for shipping charges is as described above under our Basic Limited One (1) Year Warranty.

Sealed Chamber Integrity Limited 24 Month Warranty

Princeton Instruments warrants the sealed chamber integrity of all our products for a period of twenty-four (24) months after shipment. If, at anytime within twenty-four (24) months from the date of delivery, the detector should experience a sealed chamber failure, all parts and labor needed to restore the chamber seal will be covered by us. *Open chamber products carry NO WARRANTY TO THE CCD IMAGING DEVICE, EXPRESSED OR IMPLIED.* Responsibility for shipping charges is as described above under our Basic Limited One (1) Year Warranty.

Vacuum Integrity Limited 24 Month Warranty

Princeton Instruments warrants the vacuum integrity of all our products for a period of up to twenty-four (24) months from the date of shipment. We warrant that the detector head will maintain the factory-set operating temperature without the requirement for customer pumping. Should the detector experience a Vacuum Integrity failure at anytime within twenty-four (24) months from the date of delivery all parts and labor needed to restore the vacuum integrity will be covered by us. Responsibility for shipping charges is as described above under our Basic Limited One (1) Year Warranty.

Image Intensifier Detector Limited One Year Warranty

All image intensifier products are inherently susceptible to Phosphor and/or Photocathode burn (physical damage) when exposed to high intensity light. Princeton Instruments warrants, with the exception of image intensifier products that are found to have Phosphor and/or Photocathode burn damage (which carry NO WARRANTIES EXPRESSED OR IMPLIED), all image intensifier products for a period of one (1) year after shipment. *See additional Limited One (1) year Warranty terms and conditions above, which apply to this warranty.* Responsibility for shipping charges is as described above under our Basic Limited One (1) Year Warranty.

X-Ray Detector Limited One Year Warranty

Princeton Instruments warrants, with the exception of CCD imaging device and fiber optic assembly damage due to X-rays (which carry NO WARRANTIES EXPRESSED OR IMPLIED), all X-ray products for one (1) year after shipment. *See additional Basic Limited One (1) year Warranty terms and conditions above, which apply to this warranty.* Responsibility for shipping charges is as described above under our Basic Limited One (1) Year Warranty.

Software Limited Warranty

Princeton Instruments warrants all of our manufactured software discs to be free from substantial defects in materials and / or workmanship under normal use for a period of one (1) year from shipment. Princeton Instruments does not warrant that the function of the software will meet your requirements or that operation will be uninterrupted or error free. You assume responsibility for selecting the software to achieve your intended results and for the use and results obtained from the software. In addition, during the one (1) year limited warranty. The original purchaser is entitled to receive free version upgrades. Version upgrades supplied free of charge will be in the form of a download from the Internet. Those customers who do not have access to the Internet may obtain the version upgrades on a CD-ROM from our factory for an incidental shipping and handling charge. *See Item 12 in the following section of this warranty ("Your Responsibility") for more information.*

Owner's Manual and Troubleshooting

You should read the owner's manual thoroughly before operating this product. In the unlikely event that you should encounter difficulty operating this product, the owner's manual should be consulted before contacting the Princeton Instruments technical support staff or authorized service representative for assistance. If you have consulted the owner's manual and the problem still persists, please contact the Princeton Instruments technical support staff or our authorized service representative. *See Item 12 in the following section of this warranty ("Your Responsibility") for more information.*

Your Responsibility

The above Limited Warranties are subject to the following terms and conditions:

1. You must retain your bill of sale (invoice) and present it upon request for service and repairs or provide other proof of purchase satisfactory to Princeton Instruments.
2. You must notify the Princeton Instruments factory service center within (30) days after you have taken delivery of a product or part that you believe to be defective. With the exception of customers who claim a "technical issue" with the operation of the product or part, all invoices must be paid in full in accordance with the terms of sale. Failure to pay invoices when due may result in the interruption and/or cancellation of your one (1) year limited warranty and/or any other warranty, expressed or implied.
3. All warranty service must be made by the Princeton Instruments factory or, at our option, an authorized service center.
4. Before products or parts can be returned for service you must contact the Princeton Instruments factory and receive a return authorization number (RMA). Products or parts returned for service without a return authorization evidenced by an RMA will be sent back freight collect.
5. These warranties are effective only if purchased from the Princeton Instruments factory or one of our authorized manufacturer's representatives or distributors.
6. Unless specified in the original purchase agreement, Princeton Instruments is not responsible for installation, setup, or disassembly at the customer's location.
7. Warranties extend only to defects in materials or workmanship as limited above and do not extend to any product or part which has:
 - been lost or discarded by you;
 - been damaged as a result of misuse, improper installation, faulty or inadequate maintenance or failure to follow instructions furnished by us;
 - had serial numbers removed, altered, defaced, or rendered illegible;
 - been subjected to improper or unauthorized repair; or
 - been damaged due to fire, flood, radiation, or other "acts of God" or other contingencies beyond the control of Princeton Instruments.
8. After the warranty period has expired, you may contact the Princeton Instruments factory or a Princeton Instruments-authorized representative for repair information and/or extended warranty plans.
9. Physically damaged units or units that have been modified are not acceptable for repair in or out of warranty and will be returned as received.

10. All warranties implied by state law or non-U.S. laws, including the implied warranties of merchantability and fitness for a particular purpose, are expressly limited to the duration of the limited warranties set forth above. With the exception of any warranties implied by state law or non-U.S. laws, as hereby limited, the forgoing warranty is exclusive and in lieu of all other warranties, guarantees, agreements, and similar obligations of manufacturer or seller with respect to the repair or replacement of any parts. In no event shall Princeton Instruments' liability exceed the cost of the repair or replacement of the defective product or part.
11. This limited warranty gives you specific legal rights and you may also have other rights that may vary from state to state and from country to country. Some states and countries do not allow limitations on how long an implied warranty lasts, when an action may be brought, or the exclusion or limitation of incidental or consequential damages, so the above provisions may not apply to you.
12. When contacting us for technical support or service assistance, please refer to the Princeton Instruments factory of purchase, contact your authorized Princeton Instruments representative or reseller, or visit our technical support page at www.piaction.com.

Contact Information

Roper Scientific's manufacturing facility for this product is located at the following address:

Princeton Instruments
3660 Quakerbridge Road
Trenton, NJ 08619 (USA)

Tel: 800.874.9789 / 609.587.9797

Fax: 609.587.1970

Customer Support E-mail: <mailto:techsupport@piaction.com>

For immediate support in your area, please call the following locations directly:

America	1.877.4.PIACTON (877.474.2286)
Benelux	+31 (347) 324989
France	+33 (1) 60.86.03.65
Germany	+49 (0) 89.660.7793
Japan	+81 (3) 5639.2741
UK & Ireland	+44 (0) 28.3831.0171

Otherwise, see our Support web page at www.piaction.com. An up-to-date list of addresses and telephone numbers is posted on the www.piaction.com/Support page. In addition, links on this page to support topics allow you to send e-mail based requests to the customer support group.

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